

=====

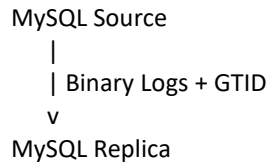
MYSQL REPLICATION USING PERCONA XTRABACKUP

SOURCE -> REPLICA

=====

PURPOSE

This runbook creates MySQL replication using:



Percona XtraBackup is used to create the initial copy of the Source database and restore it to the Replica.

This procedure assumes MySQL 8.x and GTID-based replication.

IMPORTANT:

1. Test in DEV/TEST/UAT before Production.
2. Do not execute destructive commands on an existing server until you have verified the data and backups.
3. Source and Replica should use compatible MySQL versions.
4. Server IDs must be unique.
5. Do not expose MySQL port 3306 to the public Internet.
6. Keep a backup before changing an existing server.

=====

1. ENVIRONMENT

=====

Example environment:

SOURCE / PRIMARY

Hostname : mysql-primary
IP : 10.10.10.10
MySQL : 8.4.x
Server ID: 1

REPLICA

Hostname : mysql-replica
IP : 10.10.10.20
MySQL : 8.4.x
Server ID: 2

REPLICATION USER

Username : repl
Password : VeryStrongReplicationPassword

BACKUP

Backup directory:

/backup/mysql/

IMPORTANT:

Replace the example IP addresses, hostname, username and password with your actual environment.

=====

2. CHECK MYSQL VERSION ON BOTH SERVERS

=====

Run on SOURCE:

mysql --version

mysql -uroot -p

SELECT VERSION();

Run on REPLICA:

mysql --version

mysql -uroot -p

SELECT VERSION();

Ideally:

SOURCE = MySQL 8.4.x

REPLICA = MySQL 8.4.x

Do not start replication with unsupported or incompatible versions.

=====

3. CHECK XTRABACKUP VERSION

=====

Run on SOURCE:

xtrabackup --version

Check:

mysql --version

Make sure your Percona XtraBackup version is compatible with your MySQL/Percona Server version.

```
=====
4. CONFIGURE SOURCE
=====
```

On SOURCE:

```
sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf
```

Add/check:

```
[mysqld]
```

```
server-id=1
```

```
log_bin=mysql-bin
binlog_format=ROW
```

```
gtid_mode=ON
enforce_gtid_consistency=ON
```

```
log_replica_updates=ON
```

NOTE:

For older MySQL versions, you may see:

```
log_slave_updates=ON
```

Use the setting appropriate for your MySQL version.

```
=====
5. RESTART SOURCE
=====
```

Run:

```
sudo systemctl restart mysql
```

Check:

```
sudo systemctl status mysql --no-pager
```

Check configuration:

```
mysql -uroot -p
```

Run:

```
SHOW VARIABLES LIKE 'server_id';

SHOW VARIABLES LIKE 'log_bin';

SHOW VARIABLES LIKE 'binlog_format';

SHOW VARIABLES LIKE 'gtid_mode';

SHOW VARIABLES LIKE 'enforce_gtid_consistency';
```

Expected:

```
server_id      1
log_bin        ON
binlog_format   ROW
gtid_mode       ON
enforce_gtid_consistency ON
```

```
=====
6. CREATE REPLICATION USER ON SOURCE
=====
```

Login to SOURCE:

```
mysql -uroot -p
```

Create user:

```
CREATE USER 'repl'@'10.10.10.20'
IDENTIFIED BY 'VeryStrongReplicationPassword';
```

Grant replication privilege:

```
GRANT REPLICATION SLAVE
ON *.*
TO 'repl'@'10.10.10.20';
```

Check:

```
SHOW GRANTS FOR 'repl'@'10.10.10.20';
```

IMPORTANT:

Use a strong password.

```
=====
7. CHECK MYSQL NETWORK LISTENING
=====
```

On SOURCE:

```
sudo ss -lntp | grep 3306
```

Check MySQL bind address:

```
mysql -uroot -p
```

```
SHOW VARIABLES LIKE 'bind_address';
```

The SOURCE must be reachable from the REPLICA.

```
=====
8. CONFIGURE FIREWALL ON SOURCE
=====
```

If UFW is enabled:

```
sudo ufw status
```

Allow ONLY the Replica IP:

```
sudo ufw allow from 10.10.10.20 to any port 3306 proto tcp
```

Check:

```
sudo ufw status
```

IMPORTANT:

Do NOT do:

```
sudo ufw allow 3306
```

unless you fully understand the security implications.

Only allow trusted replication hosts.

```
=====
9. TEST CONNECTION FROM REPLICA TO SOURCE
=====
```

On REPLICA:

```
mysql -h 10.10.10.10 \
-u repl \
-p \
-e "SELECT 1;"
```

Expected:

```
+---+
| 1 |
+---+
| 1 |
+---+
```

If this fails:

STOP.

Fix:

- Firewall
- Network
- bind-address
- MySQL user
- Password
- Host permission

before continuing.

```
=====
10. CHECK SOURCE BEFORE BACKUP
=====
```

On SOURCE:

```
sudo systemctl status mysql --no-pager
```

Check:

```
mysqladmin -uroot -p ping
```

Expected:

```
mysqld is alive
```

Check disk:

```
df -h
```

```
df -i
```

Check error log:

```
sudo journalctl -u mysql --since "2 hours ago"
```

Make sure there are no serious existing MySQL errors.

```
=====
11. CREATE XTRABACKUP DIRECTORY
=====
```

On SOURCE:

```
sudo mkdir -p /backup/mysql
```

Check:

```
sudo ls -lah /backup/mysql/
```

Make sure sufficient disk space exists:

```
df -h /backup
```

```
=====
12. TAKE FULL XTRABACKUP ON SOURCE
=====
```

Example:

```
sudo xtrabackup \
  --backup \
  --target-dir=/backup/mysql/full_$(date +%Y%m%d_%H%M%S)
```

Wait for:

completed OK!

Example backup:

```
/backup/mysql/full_20260819_130000/
```

DO NOT CONTINUE IF THE BACKUP DID NOT COMPLETE SUCCESSFULLY.

```
=====
13. CHECK XTRABACKUP BACKUP
=====
```

Run:

```
sudo ls -lah /backup/mysql/
```

Check size:

```
sudo du -sh /backup/mysql/*
```

Check important file:

```
sudo ls -lah \  
/backup/mysql/full_20260819_130000/xtrabackup_binlog_info
```

Read it:

```
sudo cat \  
/backup/mysql/full_20260819_130000/xtrabackup_binlog_info
```

Example:

```
mysql-bin.000005 123456 abcdef12-1234-5678-90ab-123456789abc:1-250
```

IMPORTANT:
SAVE THE GTID INFORMATION.

```
=====
```

14. UNDERSTAND xtrabackup_binlog_info

```
=====
```

Example:

```
mysql-bin.000005  
123456  
abcdef12-1234-5678-90ab-123456789abc:1-250
```

The important GTID information is:

```
abcdef12-1234-5678-90ab-123456789abc:1-250
```

You will use the GTID information when initializing the Replica.

```
=====
```

15. PREPARE THE XTRABACKUP

```
=====
```

Run:

```
sudo xtrabackup \  
--prepare \  
--target-dir=/backup/mysql/full_20260819_130000
```

Wait for:

```
completed OK!
```


Do NOT continue if prepare fails.

```
=====
16. TRANSFER BACKUP TO REPLICA
=====
```

Example using rsync:

```
sudo rsync -avP \
  /backup/mysql/full_20260819_130000/ \
  root@10.10.10.20:/backup/mysql/full_20260819_130000/
```

Or SCP:

```
sudo scp -r \
  /backup/mysql/full_20260819_130000 \
  root@10.10.10.20:/backup/mysql/
```

For large databases, rsync is generally preferable.

```
=====
17. VERIFY BACKUP ON REPLICA
=====
```

On REPLICA:

```
sudo ls -lah /backup/mysql/
```

Check:

```
sudo du -sh /backup/mysql/full_20260819_130000
```

Check:

```
sudo ls -lah \
  /backup/mysql/full_20260819_130000/xtrabackup_binlog_info
```

Read:

```
sudo cat \
  /backup/mysql/full_20260819_130000/xtrabackup_binlog_info
```

The GTID information should match the SOURCE backup.

```
=====
18. STOP MYSQL ON REPLICA
=====
```

IMPORTANT:

This step can be destructive if the Replica already contains important data.

If this is a new Replica:

```
sudo systemctl stop mysql
```

Check:

```
sudo systemctl status mysql --no-pager
```

Expected:

```
inactive / stopped
```

```
=====
19. FIND MYSQL DATA DIRECTORY
=====
```

Before moving any data:

```
mysql -uroot -p \
-e "SHOW VARIABLES LIKE 'datadir';"
```

Common result:

```
/var/lib/mysql/
```

IMPORTANT:

Verify the actual directory on your server.

```
=====
20. BACKUP EXISTING REPLICA DATA
=====
```

ONLY if the Replica already has data you need to preserve.

Example:

```
sudo mv /var/lib/mysql \
/var/lib/mysql.old_$(date +%Y%m%d_%H%M%S)
```

Then:

```
sudo mkdir /var/lib/mysql
```

```
sudo chown mysql:mysql /var/lib/mysql
```

If this is a completely new server, follow your normal initialization procedure instead.

```
=====
21. RESTORE XTRABACKUP TO REPLICA
=====
```

Run:

```
sudo xtrabackup \
  --copy-back \
  --target-dir=/backup/mysql/full_20260819_130000
```

After restore:

```
sudo chown -R mysql:mysql /var/lib/mysql
```

Check:

```
sudo ls -lah /var/lib/mysql/
```

```
=====
22. CONFIGURE REPLICA SERVER ID
=====
```

On REPLICA:

```
sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf
```

Use a DIFFERENT server ID from SOURCE:

[mysqld]

server-id=2

log_bin=mysql-bin
binlog_format=ROW

gtid_mode=ON
enforce_gtid_consistency=ON

log_replica_updates=ON

SOURCE:

server-id=1

REPLICA:

server-id=2

IMPORTANT:

Every MySQL server in the replication topology needs a unique server-id.

```
=====
23. START MYSQL ON REPLICA
=====
```

Run:

```
sudo systemctl start mysql
```

Check:

```
sudo systemctl status mysql --no-pager
```

Test:

```
mysqladmin -uroot -p ping
```

Expected:

```
mysqld is alive
```

```
=====
24. CHECK REPLICA GTID MODE
=====
```

Login:

```
mysql -uroot -p
```

Run:

```
SELECT VERSION();
```

```
SHOW VARIABLES LIKE 'server_id';
```

```
SHOW VARIABLES LIKE 'gtid_mode';
```

```
SHOW VARIABLES LIKE 'enforce_gtid_consistency';
```

Expected:

```
server_id = 2
```

```
gtid_mode = ON
```

```
enforce_gtid_consistency = ON
```

```
=====
25. INITIALIZE GTID STATE
=====
```

Get the GTID from:

```
/backup/mysql/full_20260819_130000/xtrabackup_binlog_info
```

Example:

```
abcdef12-1234-5678-90ab-123456789abc:1-250
```

For a BRAND-NEW replica with no transactions that need to be preserved, initialize the GTID state according to your MySQL version and Percona XtraBackup documentation.

For MySQL 8.4, a common clean-server workflow is:

```
RESET BINARY LOGS AND GTIDS;
```

Then:

```
SET GLOBAL gtid_purged =
'abcdef12-1234-5678-90ab-123456789abc:1-250';
```

IMPORTANT:

DO NOT RUN "RESET BINARY LOGS AND GTIDS" ON AN EXISTING SERVER WITHOUT CONFIRMING THAT ITS EXISTING GTID/BINARY LOG STATE CAN BE DISCARDED.

This command can destroy replication/binlog state.

```
=====
26. CONFIGURE SOURCE CONNECTION ON REPLICA
=====
```

On REPLICA:

```
mysql -uroot -p
```

Run:

```
CHANGE REPLICATION SOURCE TO
SOURCE_HOST='10.10.10.10',
SOURCE_PORT=3306,
SOURCE_USER='repl',
SOURCE_PASSWORD='VeryStrongReplicationPassword',
SOURCE_AUTO_POSITION=1;
```

IMPORTANT:

SOURCE_AUTO_POSITION=1 enables GTID-based auto-positioning.

```
=====
27. START REPLICATION
=====
```

Run on REPLICA:

```
START REPLICA;
```

Then:

```
SHOW REPLICA STATUS\G
```

```
=====
28. CHECK REPLICATION STATUS
=====
```

Run:

```
SHOW REPLICA STATUS\G
```

Important fields:

```
Replica_IO_Running
Replica_SQL_Running
Seconds_Behind_Source
Source_Host
Source_User
Retrieved_Gtid_Set
Executed_Gtid_Set
Last_IO_Error
Last_SQL_Error
```

Healthy example:

```
Replica_IO_Running: Yes
Replica_SQL_Running: Yes

Seconds_Behind_Source: 0

Last_IO_Error:
Last_SQL_Error:
```

If either of these is NO:

```
Replica_IO_Running: No
```

OR:

Replica_SQL_Running: No

STOP and investigate the error.

```
=====
29. TEST REPLICATION
=====
```

On SOURCE:

mysql -uroot -p

Run:

CREATE DATABASE replication_test;

Then:

USE replication_test;

Create table:

```
CREATE TABLE test_table (
  id INT PRIMARY KEY,
  name VARCHAR(100)
);
```

Insert:

```
INSERT INTO test_table
VALUES (1, 'Replication Test');
```

Check SOURCE:

```
SELECT * FROM test_table;
```

```
=====
30. CHECK DATA ON REPLICA
=====
```

On REPLICA:

mysql -uroot -p

Run:

```
SHOW DATABASES;
```

Then:

```
USE replication_test;
```

Run:

```
SELECT * FROM test_table;
```

Expected:

```
+---+-----+
| id | name      |
+---+-----+
|  1 | Replication Test |
+---+-----+
```

If the data appears:

```
REPLICATION IS WORKING.
```

```
=====
31. CHECK REPLICATION AGAIN
=====
```

On REPLICA:

```
SHOW REPLICA STATUS\G
```

Check:

```
Replica_IO_Running: Yes
```

```
Replica_SQL_Running: Yes
```

```
Last_IO_Error:
```

```
Last_SQL_Error:
```

Also check:

```
Retrieved_Gtid_Set
```

```
Executed_Gtid_Set
```

The Executed GTID set should continue to advance as transactions are committed on the SOURCE.


```
=====
32. CHECK REPLICATION ERRORS
=====
```

Run:

```
SHOW REPLICA STATUS\G
```

Check:

```
Last_IO_Error
```

```
Last_SQL_Error
```

If there is an error, get more details:

```
sudo journalctl -u mysql --since "30 minutes ago"
```

Also:

```
sudo tail -100 /var/log/mysql/error.log
```

DO NOT repeatedly run:

```
START REPLICA;
```

without understanding why replication stopped.

```
=====
33. CHECK SOURCE BINLOG
=====
```

On SOURCE:

```
mysql -uroot -p
```

Run:

```
SHOW BINARY LOG STATUS;
```

Check:

```
SHOW VARIABLES LIKE 'log_bin';
```

Expected:

```
log_bin = ON
```

```
=====
34. CHECK REPLICA BINLOG
=====
```

On REPLICA:

```
SHOW VARIABLES LIKE 'log_bin';
```

If log_bin is enabled:

```
SHOW BINARY LOG STATUS;
```

This is useful when building multi-level replication or future failover architecture.

```
=====
35. REMOVE TEST DATABASE
=====
```

After testing, on SOURCE:

```
DROP DATABASE replication_test;
```

Verify it disappears from REPLICA.

This confirms DELETE/DROP operations are also replicating.

```
=====
36. MONITOR REPLICATION
=====
```

On REPLICA:

```
SHOW REPLICA STATUS\G
```

Monitor:

```
Replica_IO_Running
Replica_SQL_Running
Seconds_Behind_Source
Last_IO_Error
Last_SQL_Error
Retrieved_Gtid_Set
Executed_Gtid_Set
```

Also monitor:

```
CPU
```

RAM
Disk
Disk I/O
MySQL errors
Replication lag

=====

37. BASIC REPLICATION TROUBLESHOOTING

=====

PROBLEM:

Replica_IO_Running = No

Check:

SHOW REPLICATION STATUS\G

Look at:

Last_IO_Error

Possible causes:

Network failure
Firewall
Replication user/password
SOURCE_HOST incorrect
SOURCE unavailable
Port 3306 blocked

PROBLEM:

Replica_SQL_Running = No

Check:

SHOW REPLICATION STATUS\G

Look at:

Last_SQL_Error

Possible causes:

SQL conflict
Missing table
Duplicate key
Data inconsistency

Unsupported statement
Corrupt data

Do not blindly skip errors.

```
=====
38. CHECK REPLICATION USER
=====
```

On SOURCE:

```
SHOW GRANTS FOR 'repl'@'10.10.10.20';
```

Make sure the user exists:

```
SELECT
  User,
  Host
FROM mysql.user
WHERE User='repl';
```

```
=====
39. CHECK NETWORK FROM REPLICA
=====
```

On REPLICA:

```
nc -vz 10.10.10.10 3306
```

Or:

```
telnet 10.10.10.10 3306
```

If unavailable, use:

```
mysql -h 10.10.10.10 -u repl -p
```

Expected:

Successful MySQL connection.

```
=====
40. IMPORTANT SECURITY RULES
=====
```

1. Do not expose port 3306 to the Internet.
2. Allow port 3306 only from trusted replication servers.

3. Use a strong replication password.
4. Do not put production passwords in shell history.
5. Do not share replication credentials.
6. Use TLS/SSL for replication where required by your security policy.
7. Keep MySQL root access restricted.

```
=====
41. IMPORTANT BACKUP RULES
=====
```

XtraBackup creates the INITIAL COPY.

After replication starts:

```
SOURCE
|
| Binary Logs
v
REPLICA
```

Replication is NOT a replacement for backups.

You should continue to maintain:

- Full XtraBackup
- Incremental XtraBackup
- Binary logs
- Off-server backup copies
- Regular restore tests

Example:

```
Full backup
+
Incremental backups
+
Binary logs
=
Point-in-time recovery strategy
```

```
=====
42. FINAL REPLICATION CHECKLIST
=====
```

SOURCE:

[] MySQL running

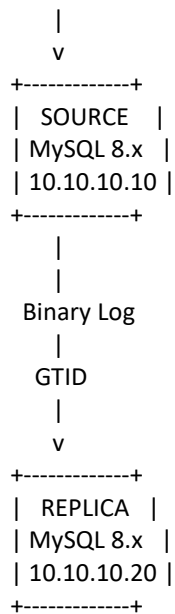
- ☐ Binary logging enabled
- ☐ GTID enabled
- ☐ binlog_format=ROW
- ☐ Unique server-id
- ☐ Replication user created
- ☐ Firewall allows Replica IP
- ☐ XtraBackup completed successfully
- ☐ xtrabackup_binlog_info saved
- ☐ Backup prepared successfully

REPLICA:

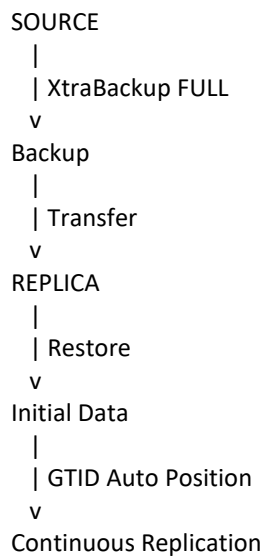
- ☐ MySQL installed
- ☐ Compatible MySQL version
- ☐ Unique server-id
- ☐ XtraBackup restored
- ☐ Correct MySQL ownership
- ☐ GTID enabled
- ☐ GTID state initialized correctly
- ☐ SOURCE connection configured
- ☐ START REPLICA executed
- ☐ Replica_IO_Running = Yes
- ☐ Replica_SQL_Running = Yes
- ☐ Last_IO_Error is empty
- ☐ Last_SQL_Error is empty
- ☐ Replication test successful
- ☐ Replication lag acceptable

=====
43. FINAL ARCHITECTURE
=====

APPLICATION



INITIAL BUILD:



44. MOST IMPORTANT COMMANDS

SOURCE:

```
mysql --version
```

```
xtrabackup --version
```

```
SHOW VARIABLES LIKE 'server_id';
```

```
SHOW VARIABLES LIKE 'log_bin';
```

```
SHOW VARIABLES LIKE 'gtid_mode';
```

SHOW BINARY LOG STATUS;

XTRABACKUP:

```
xtrabackup \  
--backup \  
--target-dir=/backup/mysql/full_YYYYMMDD_HHMMSS
```

```
xtrabackup \  
--prepare \  
--target-dir=/backup/mysql/full_YYYYMMDD_HHMMSS
```

cat /backup/mysql/full_YYYYMMDD_HHMMSS/xtrabackup_binlog_info

REPLICA:

SHOW VARIABLES LIKE 'server_id';

SHOW VARIABLES LIKE 'gtid_mode';

```
CHANGE REPLICATION SOURCE TO  
SOURCE_HOST='10.10.10.10',  
SOURCE_PORT=3306,  
SOURCE_USER='repl',  
SOURCE_PASSWORD='PASSWORD',  
SOURCE_AUTO_POSITION=1;
```

START REPLICA;

SHOW REPLICA STATUS\G

HEALTHY:

```
Replica_IO_Running: Yes  
Replica_SQL_Running: Yes  
Last_IO_Error:  
Last_SQL_Error:
```

```
=====
```

45. CHANGE RECORD

```
=====
```

SOURCE HOST:

SOURCE IP:

SOURCE MYSQL VERSION:

SOURCE SERVER ID:

REPLICA HOST:

REPLICA IP:

REPLICA MYSQL VERSION:

REPLICA SERVER ID:

XTRABACKUP VERSION:

BACKUP LOCATION:

BACKUP DATE:

SOURCE GTID:

REPLICATION START DATE:

REPLICATION START TIME:

Replica_IO_Running:

Replica_SQL_Running:

Seconds_Behind_Source:

Last_IO_Error:

Last_SQL_Error:

PERFORMED BY:

APPROVED BY:

=====
END OF MYSQL XTRABACKUP REPLICATION RUNBOOK
=====